Course Title: Experiments, A/B Testing, and Causal Inference
SDAIA Academy

Research Question & Hypothesis

QUESTION
Does a 10% discount increase purchase conversion rates among new customers?

HYPOTHESIS
Customers offered a 10% discount will convert at a higher rate.

Experimental Design

GROUP A
- Control: No discount

GROUP B
- Treatment: 10% discount

PRIMARY METRIC
- Purchase conversion rate

GUARDRAIL
- Profit per customer (not measured)

Sample Size & Duration

Planned sample: 875 per group (1,750 total)
Actual simulated allocation: A = 885, B = 865
Statistical power: 80%
Significance level: 5%
Assumed baseline: 8%
Target conversion: 12%
Estimated duration: 18 days at 100 new customers/day

Simulation Results

A · No discount



885 customers · 79 purchases

8.93%

B · 10% discount



865 customers · 107 purchases

12.37%

Difference (B – A): +3.44 percentage points

Statistical Analysis

Chi-square p-value: 0.0195 (significant at 5%)
95% confidence interval: +0.55 to +6.33 percentage points
SRM p-value: 0.6326 (no evidence of mismatch)

Conclusion & Recommendation

The discount group had higher conversion in this simulation.

These results do not establish a real-world business impact.

Run a real experiment and measure profitability before rollout.